Hill Choy

hill.choy.25@ucl.ac.uk | hillhc | +44 7871 067385, +852 6171 8745 | my website

SUMMARY

MSc student at University College London with a BEng in Computer Engineering from HKUST. Experienced in developing and optimizing Computer Vision models for real-time Edge AI applications, including traffic and pedestrian monitoring systems. Passionate about leveraging Edge AI and Computer Vision to drive operational efficiency and solve complex real-world challenges.

EDUCATION

University College London (UCL)
MSc Computer Graphics, Vision and Imaging — *Expected Graduation: September 2026*

- **Highlighted Coursework:**
  - Highlighted Coursework: Machine Vision, ML for Visual Computing, Inverse Problems, Robot Vision.

Hong Kong University of Science and Technology (HKUST)
BEng in Computer Engineering (***Second Class Honors, Division 1***) — *Graduation: July 2024*

- **Highlighted Coursework:**
  - Highlighted Coursework: Intro to NLP, Computer & Communication Networks.

IELTS Academic
8 (*Out of 9*), (*R:8.5,L:9,W:7,S:7*)

RESEARCH EXPERIENCE

Hyperspectral Image Analysis for Safe-Fabric Segmentation
UCL, MSc Industry Dissertation Project (*Partner:* ***Zori Tex***) (*Ongoing*)

- Collaborating with industry partner Zori Tex to develop an advanced CV pipeline to automate textile waste sorting and improve recycling scalability
  - Designing robust segmentation and classification algorithms to generate precise safe-fabric masks and categorize colour families. The pipeline is engineered to extract reliable features while actively mitigating the challenges of high-dimensional hyperspectral data and uneven real-world illumination.

Advanced Video Analytics and Edge AI for Smart Carpark Systems
HKUST, Bachelor Degree Final Year Project (*Supervisor:* ***Prof. Shueng-Han Gary Chan, CSE, HKUST***)

- Designed a smart car park system that utilizes edge AI to provide real-time parking space availability while minimizing computational, installation, and maintenance costs.
  - Developed a YOLOv8 model on a custom low-light/wide-angle dataset for Raspberry Pi. Enabled low-power deployment via mobile power banks by integrating a MobileNetV3 backbone and ONNX format, halving inference latency and reducing parameter count by 30%, with only a 3% drop from baseline mAP.

PROFESSIONAL EXPERIENCE

Research Assistant — **Hong Kong**
Prof. Shueng-Han Gary Chan, CSE, HKUST — *08/2024 - 06/2025*

- List of Projects:
  - K11 MUSEA Cars and Pedestrian Flow Monitoring System
    - Designed an end-to-end monitoring system for a major commercial complex, providing management with actionable real-time insights into 60,000+ daily visitor patterns to optimize mall operations and security deployment.
  - Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera
    - Developed a privacy-conserving monitoring system for elderly care that utilizes edge AI to detect falls with 95% accuracy, issuing real-time alerts that significantly reduced staff response time. Designed a web portal that automated daily reporting for 30+ patients.
    - Recipient of the **Hong Kong ICT Awards 2024, Smart People (Smart Ageing) Silver Award**.
  - Smart Traffic Management System (Kung Tong, Hong Kong)
    - Engineered real-time data capture pipelines monitoring major junctions in Kwun Tong, processing patterns for nearly 20,000 vehicles daily per junction across 11 junctions and 33 cameras to automate congestion alerts for local authorities. Responsible for the end-to-end design of data pipelines to support large-scale urban traffic management.
    - Link: https://www.police.gov.hk/

TECHNICAL SKILLS & LANGUAGES

- **Technical Stack:** Python, C++, Flask, PyTorch, YOLOv8, OpenCV, Git, Raspberry Pi, NVIDIA Jetson Nano
- **Languages**: Cantonese(Native), Mandarin(Basic), English(Fluent), Japanese(Foundation)